

Algorithm: Newton Raphson Method

1. Define Function $f(x)$
2. Define Derivative of function $f_der(x)$
3. Input:
 - a. Initial guess x_0
 - b. Tolerable Error e
4. do
 - if $f_der(x) = 0.0$ then
 - print: mathematical error
 - stop
 - end if
 - $x_1 = x_0 - f(x_0) / f_der(x_0)$
 - $x_0 = x_1$
 - $f_1 = f(x_1)$
- while ($f_1 > e$)
5. Print: root
6. Stop

Algorithm: Secant Method

1. Define Function $f(x)$
2. Input:
 - a. Initial guess x_0, x_1
 - b. Tolerable Error e
3. do
 - Evaluate $f(x_0)$ and $f(x_1)$
 - $x_2 = [x_0 * f(x_1) - x_1 * f(x_0)] / [f(x_1) - f(x_0)]$
 - $x_0 = x_1$ and $x_1 = x_2$
- while $((x_2 - x_1) / x_2) > e$,
5. Print the required root as x_2 .
6. Stop

Algorithm: Fixed Point Iterations

1. Define function as $f(x)$
2. Define convergent form $g(x)$
3. Input:
 - a. Initial guess x_0
 - b. Tolerable Error e
 - c. Maximum Iteration N
4. Initialize iteration counter: step = 1
5. Do

```
x1 = g(x0)
step = step + 1
If step > N
    Print "Not Convergent"
    Stop
End If
x0 = x1
While abs f(x1) > e
6. Print root as x1
7. Stop
```